



BUSINESS ANALYSIS MODULE

TechHub Website and WMS Outbound Process

Requirements and Implementation Pack

Customer order fulfilment across 65 stores and Main Warehouse replenishment transfers

Module	Business outcome	Primary systems	Delivery position
TH-OUT-01	Controlled, traceable customer fulfilment and store replenishment	Website • OMS services • Store network • WMS • ERP	Current-state aligned baseline for review and UAT

Purpose Define the complete outbound operating model that connects the TechHub customer website, the 65-store network and the Main Warehouse WMS without mixing customer-order stock with warehouse replenishment stock.

Primary boundary decision Customer orders allocate store order-available stock. Main Warehouse inventory is used only for separate store-replenishment transfers and becomes customer-order available after confirmed store receipt.

Contents and Document Control

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Document Control

Field	Value
Document owner	Business Analyst
Business owner	Head of Fulfilment and WMS Product Owner
Version	1.4 - Point 7 rebuilt as linked 12-field requirement records
Prepared	8 October 2026
Approval required	Digital Product Owner, Warehouse Operations Lead, Inventory Controller, Solution Architect and Integration Lead
Change control	Approved changes must update requirements, process controls, traceability and UAT evidence

1 Purpose Scope and Objectives

This pack defines the outbound process for two connected but operationally distinct flows. It provides the requirements, controls, use cases, traceability, delivery responsibilities and UAT evidence needed to validate the current TechHub website and WMS design.

1.1 Objectives

- Prevent customer orders from allocating stock held only in the Main Warehouse.
- Provide real-time store-level availability for delivery and Click and Collect across 65 stores.
- Control Main Warehouse replenishment transfers using FIFO batch allocation and location-level traceability.
- Synchronise inventory only at valid business events such as reservation, release, dispatch and store receipt.
- Provide clear operational history, shortage handling, reconciliation and audit evidence.

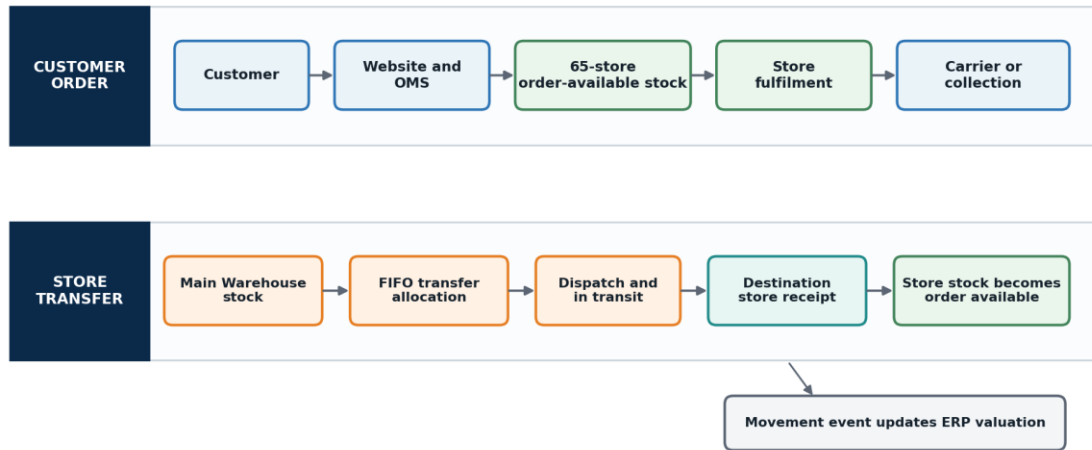
1.2 Scope boundaries

In scope	Out of scope
Customer delivery and Click and Collect order creation, store availability, reservation, fulfilment status and completion.	Supplier inbound execution, purchase-order approval and vendor payment.
Main Warehouse store-transfer creation, FIFO allocation, picking, shortage control, dispatch, in-transit status and store receipt.	Customer returns and refund processing except where an outbound status must remain traceable.
Batch, location, quantity, order type, destination, vehicle and history visibility.	Detailed carrier route optimisation and last-mile fleet management.
Website and WMS stock synchronisation and reconciliation controls.	Full ERP finance design and general-ledger configuration.

2 Operating Model and System Ownership

The operating model separates commercial order orchestration from physical warehouse execution. The website and order services control the customer promise and selected fulfilment store. The WMS controls Main Warehouse batch allocation and transfer execution. Store receipt is the hand-off that converts transferred stock into store order-available inventory.

Target system context and ownership



Boundary rule: customer orders use store stock only. Main Warehouse stock becomes order available only after destination-store receipt.

Figure 1. Target system context and ownership for website customer orders and Main Warehouse replenishment transfers.

System or team	Authoritative responsibility	Key information exchanged
Website and OMS services	Customer order, fulfilment choice, store reservation and customer-safe status	Order, store, delivery or collection choice, reservation and status
65-store network	Customer-order source stock and local fulfilment execution	Store balance, allocation, ready, dispatch or collection confirmation
Main Warehouse WMS	Warehouse inventory, batches, locations, allocations, transfer picking and dispatch	Transfer order, allocated batches, shortages, dispatch and receipt
ERP	Inventory valuation and financial record	Warehouse-to-store movement and valuation-relevant events
Carrier	Shipment handover and tracking	Shipment reference, vehicle or service and tracking events

3 Outbound Order Types and Inventory Boundaries

Order type	Source inventory	Destination	Allocation rule	Completion event
Customer delivery	Eligible store order-available stock	Customer address	Reserve one routed store; exclude Main Warehouse	Delivery confirmed
Click and Collect	Selected store order-available stock	Selected TechHub store	Complete order must be available at that store	Customer collection confirmed
Store replenishment transfer	Main Warehouse available batches	One of 65 stores	FIFO by original received batch date and approved location	Destination store receipt
Transfer cancellation	Existing Main Warehouse allocation	No physical destination	Release reservation only; do not increase physical stock	Cancelled history record

3.1 Inventory definitions

Term	Definition
Main Warehouse on hand	Physical accepted stock held in warehouse bins by SKU, batch and location.
Main Warehouse available	On hand less active transfer allocations, holds, damaged stock and other restrictions.
Store on hand	Physical stock confirmed at a branch after sale, adjustment or transfer receipt.
Store order available	Eligible store stock that can be reserved for a new website customer order.
In transit	Stock dispatched from Main Warehouse but not yet confirmed as received by the destination store.
Reserved	Stock protected for a specific customer order or store transfer and unavailable to another allocation.

4 AS IS Process

The current-state model represents the original operating problems that the TechHub solution is intended to remove: disconnected order and stock information, manual store and warehouse coordination, late shortage discovery and inconsistent status reporting.

AS-IS — Disconnected Customer Order Fulfilment

Manual store allocation creates late shortage discovery, duplicated communication and delayed customer status.

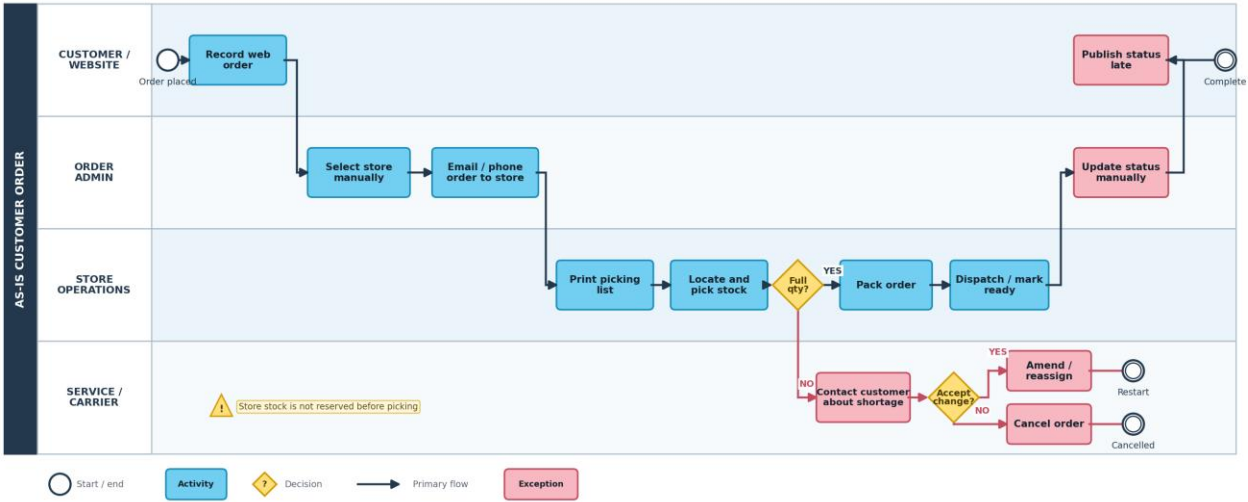


Figure 2A. Assumed current customer-order fulfilment process and exception path.

4.1 Main Warehouse store-transfer current flow

AS-IS — Manual Main Warehouse Store Transfer

Email, spreadsheets and printed tasks separate the request from physical stock and batch evidence.

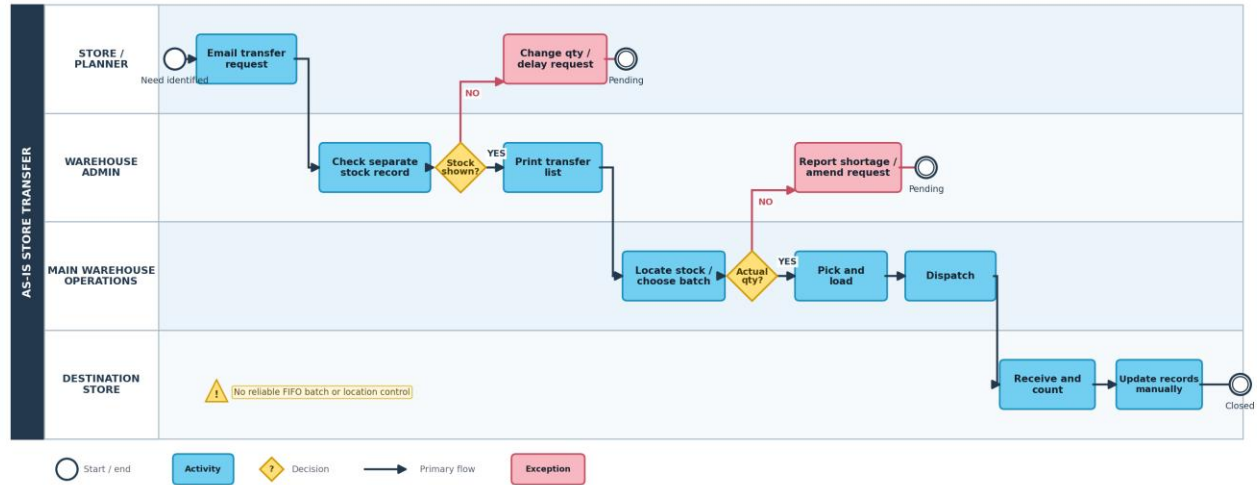


Figure 2B. Assumed current Main Warehouse store-transfer process and exception path.

Current-state issue	Business impact	Validation method
Customer and warehouse stock are not clearly separated	Overselling, incorrect promise and unexplained inventory differences	Order and inventory sample across website, stores and Main Warehouse
Store allocation relies on manual communication	Slow fulfilment and inconsistent reservation control	Store interviews and order observation
Warehouse transfer selection lacks clear FIFO batch control	Ageing batches and inconsistent picking decisions	Batch, location and dispatch history review
Shortages are discovered during or after picking	Incomplete transfers and delayed replenishment	Exception analysis and picker interviews
Status and stock corrections are re-keyed	Stale website availability and limited auditability	End-to-end order trace and reconciliation

5 TO BE Process

The target process uses two controlled outbound paths. Customer orders are fulfilled from store stock. Main Warehouse outbound work is restricted to store-replenishment transfers. Both paths publish controlled status and inventory events for synchronisation and audit.

TO-BE — Connected Customer Order Fulfilment

OMS reserves eligible stock from the 65-store network; scans, events and status updates keep the order synchronised.

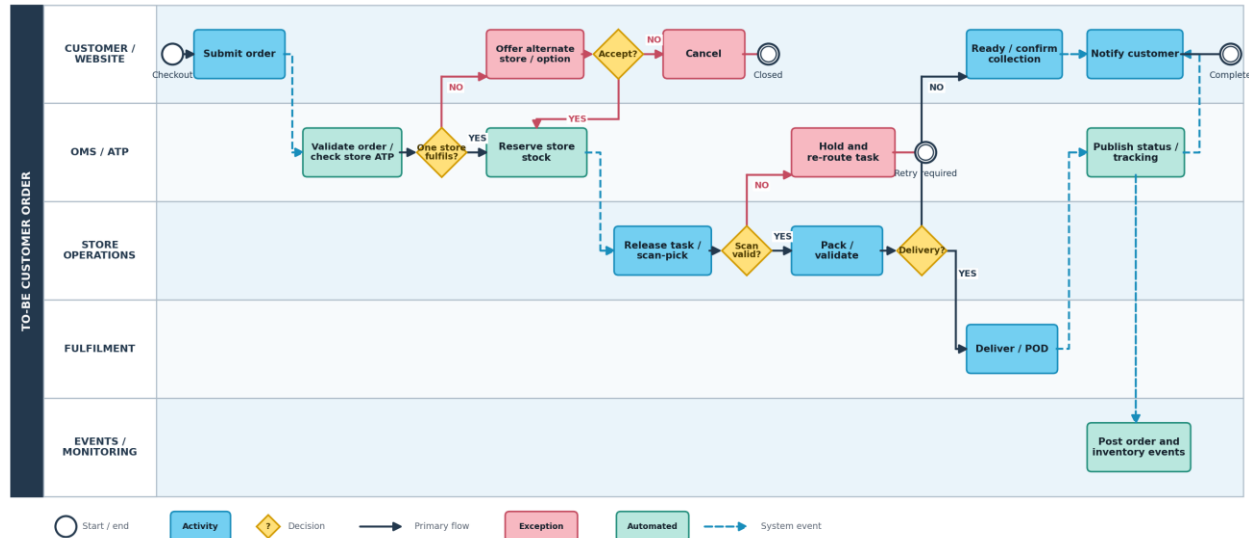


Figure 3A. Target customer-order fulfilment process with connected decisions and exception handling.

5.1 Customer-order flow

1. The customer submits the order and OMS validates the request against order-available quantities across the 65-store network.
2. If one store cannot fulfil the complete order, the website offers another store or delivery option; rejection cancels the order and acceptance returns it to reservation.
3. OMS atomically reserves the selected store stock and sends the fulfilment task to that store.
4. The store scan-picks the order. A SKU or quantity mismatch places the order on hold for correction and re-routing before picking can continue.
5. The store packs and validates the order, then follows the delivery or Click and Collect branch.
6. Delivery completes through carrier delivery and POD; Click and Collect completes after ready status and collection confirmation.
7. Fulfilment and inventory events update the order, publish tracking, notify the customer and close the order.

5.2 Main Warehouse store-transfer flow

TO-BE — Controlled Main Warehouse Store Transfer

WMS keeps Main Warehouse stock separate, allocates FIFO batches by location and releases store availability only after receipt.

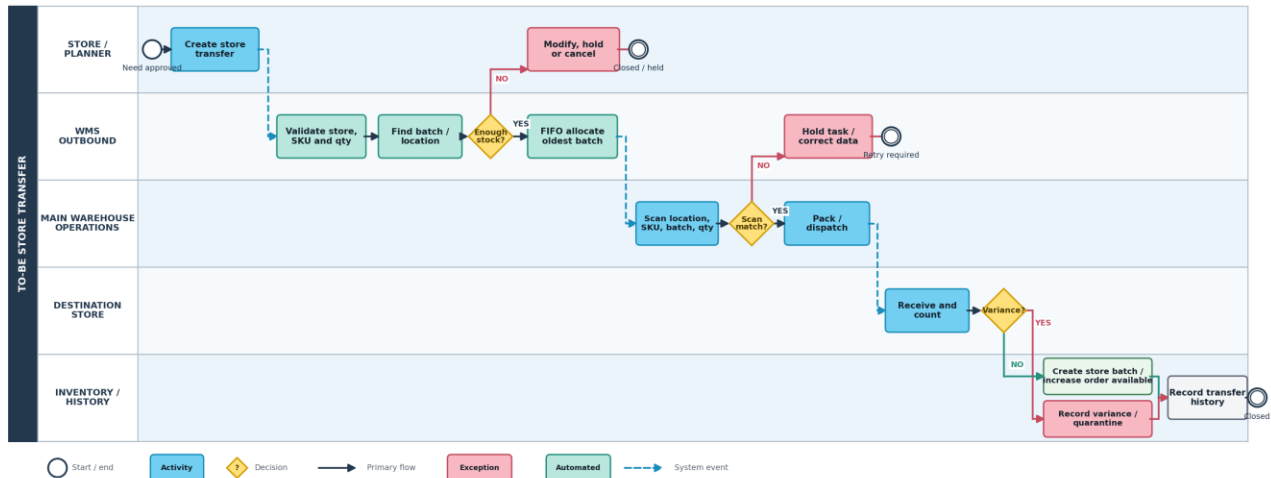
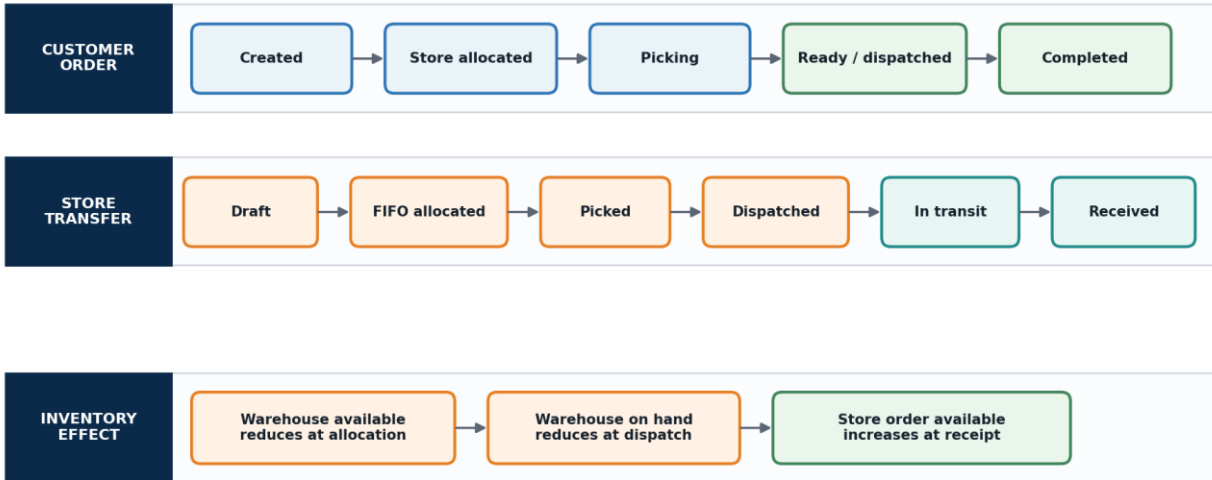


Figure 3B. Target Main Warehouse store-transfer process with FIFO, scan and receipt controls.

1. An outbound operator creates a store-replenishment order, selects the destination store and adds products by SKU or partial-name search.
2. WMS validates the destination, SKU and requested quantity, then finds eligible Main Warehouse batches and source locations.
3. If eligible stock is insufficient, WMS shows the shortage and requires the user to modify, hold or cancel the transfer.
4. If stock is sufficient, WMS allocates the oldest eligible received batch using FIFO and creates the warehouse pick task.
5. The operator scans the location, SKU, batch and quantity. A mismatch holds the task until it is corrected and successfully rescanned.
6. Confirmed stock is packed and dispatched; Main Warehouse on hand reduces and the transfer becomes in transit.
7. The destination store receives and counts the transfer, then records any shortage, overage or damage for quarantine and investigation.
8. A clean receipt creates the store batch using the original received date and increases store on hand and order-available stock.
9. WMS records the movements, status events and outbound history before closing the transfer.

6 Business Rules and Controls

Outbound status model and inventory effect



Allocation changes availability; dispatch changes Main Warehouse on hand; confirmed receipt changes store availability.

Figure 4. Outbound status transitions and controlled inventory effects.

Control point	Rule	Owner
Customer allocation	Use eligible store order-available stock only; never use Main Warehouse stock directly.	Digital Product Owner
Order completeness	Click and Collect is accepted only when one selected store can supply the complete order.	Store Operations
Transfer destination	A Main Warehouse outbound transfer must name one active destination store.	Warehouse Operations
Batch allocation	Allocate oldest eligible received batch first and retain its original received date.	Inventory Control
Location control	One SKU and batch combination is held in one approved Main Warehouse location.	Inventory Control
Shortage control	Show requested, allocated and shortage quantity; highlight shortage in red.	Warehouse Supervisor
Dispatch control	Only fully allocated and confirmed lines may dispatch.	Dispatch Lead
Store receipt	Receipt moves in-transit stock to store on hand and order available exactly once.	Store Operations
Synchronisation	Use idempotent events and reconcile balances after every controlled stock event.	Integration Lead

7. Requirement catalogue

7.1 General requirements - Part A

Requirement ID	Type	Requirement statement	Acceptance criteria
GEN-OUT-001	General	The solution shall maintain separate outbound flows for customer orders and Main Warehouse store-replenishment transfers.	Users can identify the order type, source facility, destination and inventory effect throughout the transaction.
GEN-OUT-002	General	The outbound process shall maintain traceability from order creation through allocation, picking, dispatch, receipt and completion.	An authorised user can trace the order, item, source, destination, batch, location, event, actor and status history.
GEN-OUT-003	General	Main Warehouse stock shall remain separate from store order-available stock and customer-order allocation.	Customer allocation excludes Main Warehouse inventory until an authorised transfer is received and promoted at a store.
GEN-OUT-004	General	All outbound transactions shall prevent negative physical, reserved, available and in-transit balances.	Allocation, release, cancellation, dispatch and receipt do not produce a negative balance or duplicate stock movement.
GEN-OUT-005	General	Outbound actions and status transitions shall be controlled by authorised roles and approved process rules.	Unauthorised actions are blocked, approved transitions are enforced and sensitive overrides retain user, time and reason.

General requirements - Part B: governance metadata

Requirement ID	Priority	Business area	Requester	Owner	Author / date	Estimate	Sprint / phase
GEN-OUT-001	Must	Outbound Operations	Product Sponsor	WMS Product Owner	BA 8 Oct 2026	Cross-cutting	Phase 1
GEN-OUT-002	Must	Outbound Operations	Operations Sponsor	Warehouse Operations Lead	BA 8 Oct 2026	Cross-cutting	Phase 1
GEN-OUT-003	Must	Inventory Control	Inventory Control	Inventory Control Lead	BA 8 Oct 2026	Cross-cutting	Phase 1
GEN-OUT-004	Must	Inventory Control	Operations Sponsor	WMS Product Owner	BA 8 Oct 2026	Cross-cutting	Phase 2
GEN-OUT-005	Must	Governance / Security	Risk and Operations	Product Owner	BA 8 Oct 2026	Cross-cutting	Phase 2

7.2 Functional requirements - Part A

Requirement ID	Type	Requirement statement	Acceptance criteria
FR-OUT-001	Functional	The website shall validate complete-order availability across the 65-store network before accepting a customer fulfilment choice.	Checkout identifies one eligible source store for the complete order or presents a controlled alternative without creating a reservation.
FR-OUT-002	Functional	Customer orders shall reserve stock only at the selected or routed source store.	Reservation reduces available quantity once at the source store and leaves Main Warehouse stock unchanged.
FR-OUT-003	Functional	The website shall support delivery and Click and Collect customer-order types.	Exactly one fulfilment type, source store and delivery destination or pickup store is recorded on the order.
FR-OUT-004	Functional	The website shall present customer-safe fulfilment statuses derived from approved operational events.	Processing, ready for pickup or delivery, dispatched, completed, cancelled and refunded statuses map to valid internal events.
FR-OUT-005	Functional	WMS shall allow an authorised user to create a Main Warehouse replenishment transfer to any active destination among the 65 stores.	A transfer cannot be saved unless its source is Main Warehouse and its destination is an active store.
FR-OUT-006	Functional	Outbound operators shall add transfer items by partial SKU or product-name search and enter the required quantity.	Case-insensitive partial search returns active matching items and supports quantity change and line removal.
FR-OUT-007	Functional	WMS shall validate order type, source, destination, product status and requested quantity before saving a transfer.	Invalid or missing values are identified with actionable messages and no inventory balance is changed.
FR-OUT-008	Functional	WMS shall record a unique transfer reference, service priority, requested date, carrier or vehicle and transfer notes.	The saved transfer displays all mandatory shipment metadata and retains it through dispatch and history.
FR-OUT-009	Functional	WMS shall allocate transfer quantities from eligible Main Warehouse stock using FIFO by original received batch date.	The oldest eligible received batch is allocated first and the original received date is retained through store receipt.
FR-OUT-010	Functional	WMS shall enforce the approved rule that one SKU and batch combination occupies one Main Warehouse storage location.	Allocation identifies the approved location and prevents a conflicting active location assignment for the same SKU and batch.
FR-OUT-011	Functional	WMS shall display requested, allocated and remaining quantities together with batch and source location on each transfer line.	Before picking, users can view the exact quantity, batch and location selected for every allocated line.
FR-OUT-012	Functional	WMS shall calculate shortage quantity when eligible Main Warehouse stock cannot satisfy a transfer line.	The shortage equals requested less allocated quantity, is visibly highlighted and blocks dispatch until resolved.

Requirement ID	Type	Requirement statement	Acceptance criteria
FR-OUT-013	Functional	An authorised supervisor shall be able to hold, amend or cancel an unresolved transfer before dispatch.	The action requires a reason, follows status rules and releases any active allocation exactly once when applicable.
FR-OUT-014	Functional	Picking confirmation shall validate the allocated source location, SKU, batch and quantity.	A mismatched, blocked or unallocated scan is rejected and the picker receives a clear corrective message.
FR-OUT-015	Functional	WMS shall record short picks and other picking exceptions against the affected transfer line.	The user records quantity and reason, the remaining shortage is visible and dispatch stays blocked until an approved resolution.
FR-OUT-016	Functional	Dispatch confirmation shall be allowed only for a fully allocated and confirmed transfer.	Dispatch consumes reservations once, reduces Main Warehouse on hand, increases in-transit quantity and changes status to in transit.
FR-OUT-017	Functional	WMS shall produce a dispatch record containing transfer, destination, carrier or vehicle, item, quantity, batch and tracking details.	The dispatch record can be viewed or printed and its reference remains linked to the transfer history.
FR-OUT-018	Functional	The destination store shall confirm receipt of a dispatched transfer and record the counted quantity.	A valid receipt reduces in-transit quantity and increases store on hand exactly once.
FR-OUT-019	Functional	The destination store shall record shortage, overage or damage identified during receipt.	Each variance records item, expected quantity, counted quantity, condition, reason and resolution status without silently changing the expected quantity.
FR-OUT-020	Functional	Eligible stock received at a destination store shall become order available only after confirmed receipt and required status validation.	A clean receipt promotes eligible quantity once, while damaged, quarantined or unresolved quantity remains unavailable for customer orders.
FR-OUT-021	Functional	WMS shall provide active outbound and outbound-history views with operational filters.	Authorised users can filter by order type, reference, status, source, destination store, SKU, batch, date, vehicle and tracking number.
FR-OUT-022	Functional	The solution shall restrict transfer creation, allocation, picking, dispatch, cancellation, receipt and override actions by role.	Each role can perform only its approved actions and every denied attempt is rejected without changing business data.
FR-OUT-023	Functional	Website, OMS, WMS and store balances shall synchronise after reservation, release, cancellation, dispatch and receipt events.	The affected source, destination, reserved, available and in-transit balances reconcile without duplicate movement events.
FR-OUT-024	Functional	Authorised support users shall be able to identify failed outbound events, retry recoverable events and reconcile affected balances.	Failed events appear in an exception view with reason and status; an approved retry completes once without duplicating a transaction.
FR-OUT-025	Functional	Authorised users shall be able to export filtered outbound history for operational review and audit.	The export respects the active filters and includes reference, type, status, facilities, item, batch, quantity, actor and timestamps.

Functional requirements - Part B: governance metadata

Requirement ID	Priority	Business area	Requester	Owner	Author / date	Estimate	Sprint / phase
FR-OUT-001	Must	Digital Commerce	Customer Experience	Digital Product Owner	BA 8 Oct 2026	8 SP	Sprint 1
FR-OUT-002	Must	Order Management	Customer Experience	OMS Owner	BA 8 Oct 2026	8 SP	Sprint 1
FR-OUT-003	Must	Digital Commerce	Customer Experience	Digital Product Owner	BA 8 Oct 2026	5 SP	Sprint 1
FR-OUT-004	Must	Digital Commerce	Customer Experience	Digital Product Owner	BA 8 Oct 2026	5 SP	Sprint 2
FR-OUT-005	Must	Warehouse Outbound	Warehouse Operations	WMS Product Owner	BA 8 Oct 2026	8 SP	Sprint 2
FR-OUT-006	Should	Warehouse Outbound	Warehouse Operations	WMS Product Owner	BA 8 Oct 2026	5 SP	Sprint 2
FR-OUT-007	Must	Warehouse Outbound	Warehouse Operations	WMS Product Owner	BA 8 Oct 2026	5 SP	Sprint 2
FR-OUT-008	Must	Warehouse Outbound	Transport Operations	Warehouse Operations Lead	BA 8 Oct 2026	5 SP	Sprint 3
FR-OUT-009	Must	Inventory Control	Inventory Control	Inventory Control Lead	BA 8 Oct 2026	13 SP	Sprint 3
FR-OUT-010	Must	Inventory Control	Inventory Control	Inventory Control Lead	BA 8 Oct 2026	8 SP	Sprint 3
FR-OUT-011	Must	Inventory Control	Warehouse Operations	WMS Product Owner	BA 8 Oct 2026	5 SP	Sprint 3
FR-OUT-012	Must	Inventory Control	Warehouse Operations	WMS Product Owner	BA 8 Oct 2026	5 SP	Sprint 3
FR-OUT-013	Must	Warehouse Outbound	Warehouse Operations	Warehouse Operations Lead	BA 8 Oct 2026	8 SP	Sprint 4
FR-OUT-014	Must	Warehouse Outbound	Picking Team	Picking Lead	BA 8 Oct 2026	8 SP	Sprint 4
FR-OUT-015	Should	Warehouse Outbound	Picking Team	Picking Lead	BA 8 Oct 2026	5 SP	Sprint 4
FR-OUT-016	Must	Warehouse Outbound	Dispatch Team	Dispatch Lead	BA 8 Oct 2026	13 SP	Sprint 4
FR-OUT-017	Should	Warehouse Outbound	Transport Operations	Dispatch Lead	BA 8 Oct 2026	5 SP	Sprint 4
FR-OUT-018	Must	Store Operations	Store Operations	Store Operations Lead	BA 8 Oct 2026	8 SP	Sprint 5

Requirement ID	Priority	Business area	Requester	Owner	Author / date	Estimate	Sprint / phase
FR-OUT-019	Must	Store Operations	Store Operations	Store Operations Lead	BA 8 Oct 2026	8 SP	Sprint 5
FR-OUT-020	Must	Store Inventory	Store Operations	Inventory Control Lead	BA 8 Oct 2026	8 SP	Sprint 5
FR-OUT-021	Must	Warehouse Outbound	Warehouse Operations	Warehouse Operations Lead	BA 8 Oct 2026	8 SP	Sprint 5
FR-OUT-022	Must	Governance / Security	Risk and Operations	Security Owner	BA 8 Oct 2026	8 SP	Sprint 5
FR-OUT-023	Must	Integration	IT Operations	Integration Lead	BA 8 Oct 2026	13 SP	Sprint 6
FR-OUT-024	Must	Integration Support	IT Operations	Support Lead	BA 8 Oct 2026	8 SP	Sprint 6
FR-OUT-025	Should	Reporting / Audit	Operations and Audit	Data Lead	BA 8 Oct 2026	5 SP	Sprint 6

7.3 Non-functional requirements - Part A

Requirement ID	Type	Requirement statement	Acceptance criteria
NFR-OUT-001	Non-functional	Confirmed outbound inventory and status events shall appear in relevant operational views within the approved synchronisation target.	At least 95 percent of controlled pilot events meet the approved end-to-end synchronisation target.
NFR-OUT-002	Non-functional	Duplicate order, allocation, release, dispatch and receipt requests shall be processed idempotently.	Replaying the same business request does not duplicate an order, allocation, status change or stock movement.
NFR-OUT-003	Non-functional	The solution shall retain a complete timestamped and tamper-evident audit trail for outbound activity.	Each controlled change records reference, event, actor or system, facility, before and after status and quantity effect.
NFR-OUT-004	Non-functional	Transfer search, line validation and operational history queries shall respond within the approved performance targets under normal pilot load.	At least 95 percent of controlled interactions meet the approved response-time target without data loss.
NFR-OUT-005	Non-functional	Core outbound creation, allocation, picking, dispatch and receipt functions shall meet the approved operational availability target.	Monitoring demonstrates the approved availability target during agreed warehouse and store operating hours.
NFR-OUT-006	Non-functional	Integration failures shall be visible, recoverable and reconcilable without losing a confirmed business transaction.	Failed events enter a monitored retry or exception path and recovery evidence confirms consistent final balances.
NFR-OUT-007	Non-functional	Outbound data and actions shall be protected by authenticated access, least-privilege roles and approved session controls.	Access tests prove that authorised roles can perform approved actions and unauthorised users cannot view or change restricted data.

Requirement ID	Type	Requirement statement	Acceptance criteria
NFR-OUT-008	Non-functional	The outbound solution shall support the Main Warehouse and all 65 stores at the approved peak transaction and user volumes.	Load testing meets approved throughput, concurrency and error-rate targets for the pilot and rollout volumes.

Non-functional requirements - Part B: governance metadata

Requirement ID	Priority	Business area	Requester	Owner	Author / date	Estimate	Sprint / phase
NFR-OUT-001	Must	Technology / Operations	IT Operations	Solution Architect	BA 8 Oct 2026	5 SP	Sprint 5
NFR-OUT-002	Must	Integration	Enterprise Architecture	Integration Lead	BA 8 Oct 2026	8 SP	Sprint 4
NFR-OUT-003	Must	Risk / Audit	Operations and Audit	Support Lead	BA 8 Oct 2026	5 SP	Sprint 5
NFR-OUT-004	Must	Technology / Operations	Warehouse Operations	Solution Architect	BA 8 Oct 2026	8 SP	Sprint 5
NFR-OUT-005	Must	Technology / Operations	IT Operations	Support Lead	BA 8 Oct 2026	8 SP	Sprint 6
NFR-OUT-006	Must	Integration	IT Operations	Integration Lead	BA 8 Oct 2026	8 SP	Sprint 6
NFR-OUT-007	Must	Security	Information Security	Security Owner	BA 8 Oct 2026	8 SP	Sprint 5
NFR-OUT-008	Should	Technology / Operations	Operations Sponsor	Solution Architect	BA 8 Oct 2026	13 SP	Sprint 6

7.4 Technical requirements - Part A

Requirement ID	Type	Requirement statement	Acceptance criteria
TR-OUT-001	Technical	Every customer order, transfer, allocation, shipment and movement shall use a unique immutable business reference and correlation identifier.	References remain consistent across interfaces, operational views, history and exports.
TR-OUT-002	Technical	Stock changes shall be posted as controlled movement events rather than direct balance overwrites.	Reservation, release, dispatch, receipt and adjustment each create one attributable quantity-effect event.
TR-OUT-003	Technical	Concurrency control shall prevent the same eligible stock from being allocated more than once.	Parallel allocation attempts cannot reserve more than the eligible available quantity and one request receives a controlled conflict response.
TR-OUT-004	Technical	Outbound APIs and services shall validate mandatory fields, business rules and status transitions and return actionable errors.	Invalid order type, store, item, quantity, batch, location or transition is rejected without changing stock.
TR-OUT-005	Technical	Batch records shall retain the original received date and use it as the FIFO priority value across warehouse-to-store transfer.	Store receipt preserves the source batch date rather than creating a newer ageing date.

Requirement ID	Type	Requirement statement	Acceptance criteria
TR-OUT-006	Technical	Inventory balances shall be reproducible from receipts, reservations, releases, dispatches, receipts and authorised adjustments.	Reconciliation reproduces on-hand, reserved, available and in-transit balances from the movement ledger.
TR-OUT-007	Technical	Outbound integration shall provide controlled retry, dead-letter handling and authorised replay for failed events.	A recoverable failure can be retried or replayed once using the original identifier without duplicating business effects.
TR-OUT-008	Technical	Operational monitoring shall correlate application, integration and inventory events using the same business reference.	Support users can trace a transaction across components and alerts contain reference, failure point, time and severity.
TR-OUT-009	Technical	Outbound services shall enforce authenticated identity and role claims for every restricted command.	Interface tests reject missing, expired or unauthorised credentials and record the denied attempt.
TR-OUT-010	Technical	Outbound interfaces shall use approved versioned contracts, encrypted transport and controlled schema compatibility.	Contract and security tests validate required fields, schema version, compatibility rules and encrypted transport.

Technical requirements - Part B: governance metadata

Requirement ID	Priority	Business area	Requester	Owner	Author / date	Estimate	Sprint / phase
TR-OUT-001	Must	Integration / Architecture	Enterprise Architecture	Integration Lead	BA 8 Oct 2026	5 SP	Sprint 2
TR-OUT-002	Must	Integration / Architecture	Enterprise Architecture	Solution Architect	BA 8 Oct 2026	8 SP	Sprint 3
TR-OUT-003	Must	WMS Engineering	Inventory Control	Technical Lead	BA 8 Oct 2026	8 SP	Sprint 3
TR-OUT-004	Must	Integration / Architecture	Enterprise Architecture	Technical Lead	BA 8 Oct 2026	8 SP	Sprint 3
TR-OUT-005	Must	WMS Engineering	Inventory Control	WMS Product Owner	BA 8 Oct 2026	5 SP	Sprint 3
TR-OUT-006	Must	Data / Integration	Finance and Inventory	Data Lead	BA 8 Oct 2026	8 SP	Sprint 5
TR-OUT-007	Must	Integration / Architecture	IT Operations	Integration Lead	BA 8 Oct 2026	8 SP	Sprint 6
TR-OUT-008	Must	Technology Operations	IT Operations	Support Lead	BA 8 Oct 2026	5 SP	Sprint 6
TR-OUT-009	Must	Security / Architecture	Information Security	Security Owner	BA 8 Oct 2026	8 SP	Sprint 5
TR-OUT-010	Must	Integration / Architecture	Enterprise Architecture	Solution Architect	BA 8 Oct 2026	8 SP	Sprint 4

8 Use Cases

The use-case boundary shows the operational actions performed by customers, store users and warehouse users, together with the supporting carrier, supervisor and ERP interactions.

Outbound use cases by actor

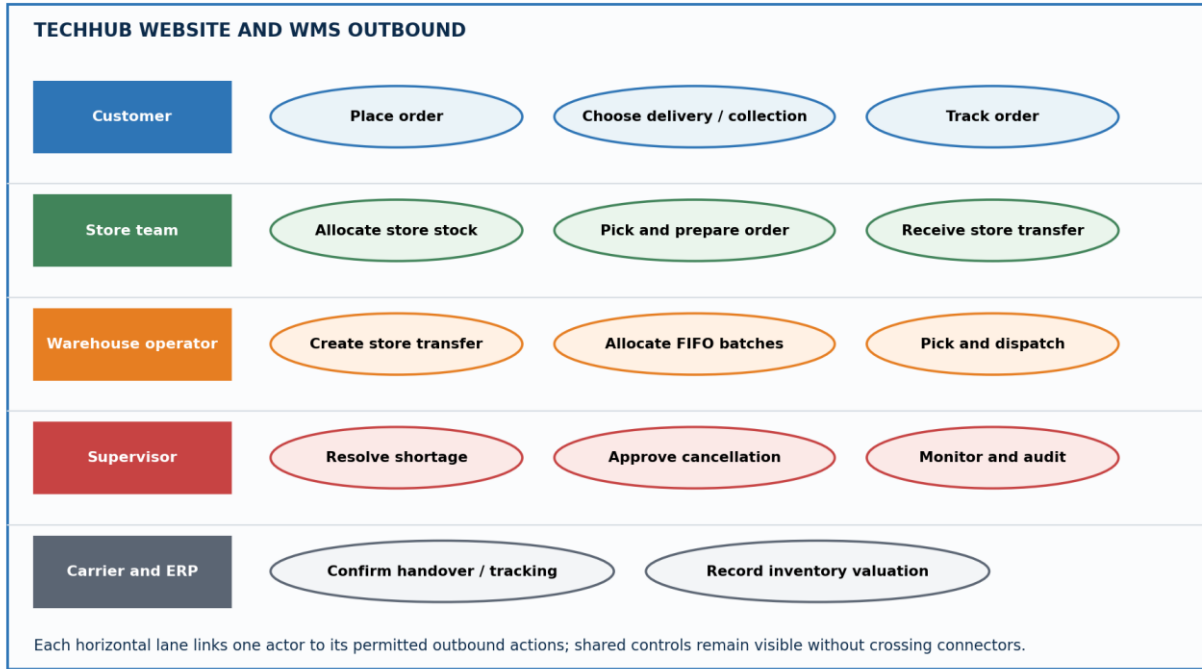


Figure 5. Website and WMS outbound use-case boundary.

ID	Use case	Primary actor	Precondition	Success outcome	Exception
UC-OUT-01	Place and track customer order	Customer	Eligible store order-available stock exists	Order is reserved once and customer-safe status is visible	No single eligible store can fulfil the request
UC-OUT-02	Allocate store customer order	Store team or order service	Accepted order with source store	Source-store quantity is reserved once	Concurrent request or stock restriction
UC-OUT-03	Create store replenishment transfer	Outbound operator	Authorised user, active store and catalogue items	Valid transfer, shipment metadata and requested lines are recorded	Invalid role, store, item, quantity or mandatory field
UC-OUT-04	Allocate FIFO warehouse batches	WMS	Eligible Main Warehouse stock exists	Oldest batches and approved source locations are reserved	Shortage, blocked stock or location conflict
UC-OUT-05	Pick and dispatch transfer	Warehouse operator	Fully allocated transfer	Correct stock is confirmed, dispatch record is created and transfer is in transit	Wrong scan, short pick or incomplete control
UC-OUT-06	Receive transfer at store	Store team	Transfer is in transit to destination store	Receipt is recorded and eligible stock becomes order available	Duplicate receipt, shortage, overage or damage

9 User Stories and Acceptance Criteria

Each story is linked to the outbound requirement catalogue. Acceptance criteria are observable outcomes that can be converted into SIT and UAT test cases during refinement.

US-OUT-001 - Customer fulfilment source

Related requirements: GEN-OUT-001-004; FR-OUT-001-004; FR-OUT-023

As a customer, I want the website to validate eligible store stock and fulfilment options so that my order can be confirmed without using Main Warehouse stock.

- **The complete order is sourced from eligible store order-available stock.**
- **Delivery or Click and Collect records the source store and fulfilment destination.**
- **If no store can fulfil the complete order, the customer is offered an alternative and no invalid reservation is created.**

US-OUT-002 - Store order allocation

Related requirements: FR-OUT-002; NFR-OUT-002; TR-OUT-003

As an order service owner, I want customer stock reserved once at the selected store so that concurrent orders cannot over-allocate the same quantity.

- **A confirmed order reduces available quantity only at its selected or routed store.**
- **Main Warehouse inventory remains unchanged by customer-order allocation.**
- **Concurrent or repeated requests cannot reserve more than the eligible store quantity.**

US-OUT-003 - Store replenishment transfer creation

Related requirements: FR-OUT-005-008; TR-OUT-001; TR-OUT-004; TR-OUT-009

As an outbound operator, I want to create a Main Warehouse replenishment transfer for any active store so that required products can be dispatched through a controlled order.

- **The transfer records a unique reference, destination store, carrier or vehicle, priority and requested lines.**
- **Product search accepts a partial SKU or product name and returns active matching items.**
- **Invalid stores, products, quantities or mandatory fields are rejected without changing stock.**

US-OUT-004 - FIFO batch and location allocation

Related requirements: FR-OUT-009-012; TR-OUT-003; TR-OUT-005-006

As an inventory controller, I want transfer quantities allocated from the oldest eligible Main Warehouse batches so that FIFO and location traceability are maintained.

- **The oldest eligible received batch is allocated first using its original received date.**
- **Each allocated line shows quantity, batch and approved source location.**
- **Shortage quantities are shown clearly and dispatch remains blocked until the shortage is resolved.**

US-OUT-005 - Picking and dispatch validation

Related requirements: FR-OUT-014-017; TR-OUT-002; TR-OUT-004

As a warehouse operator, I want picking and dispatch validated against the allocated location, SKU, batch and quantity so that only correct stock leaves the Main Warehouse.

- **A mismatched or blocked scan is rejected with a clear corrective message.**
- **Only fully allocated and confirmed transfer lines can be dispatched.**
- **Dispatch posts one controlled movement, reduces Main Warehouse on hand and changes the transfer to in transit.**

US-OUT-006 - Transfer hold or cancellation

Related requirements: GEN-OUT-004-005; FR-OUT-012-013; FR-OUT-022; TR-OUT-002

As a warehouse supervisor, I want to hold, amend or cancel an unresolved transfer so that shortages and invalid requests do not create incorrect stock balances.

- **A shortage can be held for correction or returned for an approved quantity change.**
- **Cancellation releases any active allocation exactly once and does not increase physical stock.**
- **The reason, user, time and resulting inventory effect remain visible in history.**

US-OUT-007 - Destination store receipt

Related requirements: FR-OUT-018-020; TR-OUT-002; TR-OUT-005-006

As a destination store user, I want to receive and count a dispatched transfer so that store stock becomes order available only after confirmed receipt.

- **Receipt reduces in-transit quantity and increases store on hand exactly once.**
- **A clean receipt creates or updates store stock while preserving the original batch received date.**
- **Shortage, overage or damage is recorded as a variance and does not silently change the expected quantity.**

US-OUT-008 - Outbound history and audit

Related requirements: FR-OUT-021; FR-OUT-025; NFR-OUT-003; TR-OUT-001; TR-OUT-008

As an authorised operations user, I want searchable outbound history so that I can trace each customer order or store transfer from creation to completion.

- **Users can filter by order type, status, destination store, SKU, batch, date and vehicle.**
- **History records the reference, event, status, actor or system, facility, timestamp and quantity effect.**
- **The same immutable reference is retained across operational views and exports.**

US-OUT-009 - Synchronisation and recovery

Related requirements: FR-OUT-023-024; NFR-OUT-001-002; NFR-OUT-006; TR-OUT-002; TR-OUT-006-008

As a support analyst, I want website and WMS stock events synchronised and recoverable so that confirmed business transactions remain accurate during retries or integration failures.

- **Reservation, release, dispatch and receipt update the affected availability views within the agreed interval.**
- **Replayed events do not duplicate orders, allocations or stock movements.**
- **Failed events enter a monitored retry path and reconciliation reproduces on-hand, reserved, available and in-transit balances.**

10 Requirements Traceability

Epic	Requirement(s)	Process	Use case	User story	Verification
EP-OUT-CUS-01	GEN-OUT-001-004; FR-OUT-001-004; FR-OUT-023	Customer-order fulfilment	UC-OUT-01-02	US-OUT-001-002	SIT, integration and website UAT
EP-OUT-TRF-01	FR-OUT-005-008; FR-OUT-022; TR-OUT-001; TR-OUT-004; TR-OUT-009	Transfer creation	UC-OUT-03	US-OUT-003	WMS functional, access and UAT
EP-OUT-INV-01	GEN-OUT-003-004; FR-OUT-009-013; TR-OUT-002-006	FIFO allocation and shortage	UC-OUT-04	US-OUT-004; US-OUT-006	Allocation, concurrency and reconciliation
EP-OUT-EXE-01	FR-OUT-014-017; NFR-OUT-003-004; TR-OUT-002; TR-OUT-004	Pick and dispatch	UC-OUT-05	US-OUT-005	WMS UAT, scan and dispatch evidence
EP-OUT-REC-01	FR-OUT-018-020; TR-OUT-002; TR-OUT-005-006	Store receipt and promotion	UC-OUT-06	US-OUT-007	Receipt, variance and inventory UAT
EP-OUT-AUD-01	FR-OUT-021; FR-OUT-025; NFR-OUT-003; TR-OUT-001; TR-OUT-008	History and audit	Operational review	US-OUT-008	History, export and audit evidence
EP-OUT-SEC-01	GEN-OUT-005; FR-OUT-022; NFR-OUT-007; TR-OUT-009	Role and access control	All use cases	US-OUT-003; US-OUT-006-008	Role matrix and negative access tests
EP-OUT-INT-01	FR-OUT-023-024; NFR-OUT-001-002; NFR-OUT-005-006; NFR-OUT-008; TR-OUT-006-008; TR-OUT-010	Synchronisation and recovery	Operational control	US-OUT-009	Performance, recovery and reconciliation

Traceability rule: any change to a requirement must trigger review of its linked process step, control, use case, UAT case and deployment evidence.

11 Delivery Plan and RACI

11.1 Work breakdown and indicative sequence

Phase	Key outputs	Indicative timing	Exit evidence
1 Validate and design	Confirm order types, stock definitions, process controls, batch rules and interfaces	3 weeks	Approved process and design
2 Build and configure	Website store reservation plus WMS transfer, FIFO, dispatch, receipt and history	4 to 6 weeks	Build complete and reviewed
3 Test and pilot	SIT, reconciliation and UAT with selected stores and warehouse users	3 weeks	Business acceptance and pilot approval
4 Controlled rollout	Enable stores in groups with monitoring, support and rollback readiness	2 to 4 weeks	Stable operation and support handover

11.2 RACI summary

Deliverable	Sponsor	Product owner	BA	Warehouse lead	Architect and integration	QA and SMEs
Scope and success measures	A	R	R	C	C	C
AS IS and TO BE process	I	A	R	C	C	C
Requirement baseline	I	A	R	C	C	C
Solution and interfaces	I	A	C	C	R	C
SIT and reconciliation	I	A	C	C	R	R
UAT and readiness	I	A	R	R	C	R
Pilot evidence gate	A	R	C	C	C	C

R = Responsible A = Accountable C = Consulted I = Informed

12 UAT Test Cases and Evidence Gate

UAT validates the public website order flow, Main Warehouse replenishment transfer flow and inventory synchronisation boundary. Evidence should include screenshots, order references, movement records, before-and-after balances and defect references where applicable.

Test ID	Scenario	Precondition / action	Expected result	Requirement	Status	Comment
UAT-OUT-001	Customer delivery with sufficient store stock	Eligible store can fulfil complete order; Place a delivery order	One source store is reserved once; Main Warehouse is unchanged	FR-OUT-001-003		
UAT-OUT-002	Click and Collect at selected store	Selected store can fulfil complete order; Select store and submit checkout	Order records pickup store and reserves its eligible stock once	FR-OUT-001-003		
UAT-OUT-003	No store can fulfil complete order	No single store has all requested items; Attempt checkout	Customer receives a controlled alternative and no reservation is created	FR-OUT-001		
UAT-OUT-004	Concurrent customer reservation	One remaining eligible unit exists; Submit competing orders	Only one reservation succeeds and no balance becomes negative	GEN-OUT-004; FR-OUT-002; TR-OUT-003		
UAT-OUT-005	Create Main Warehouse store transfer	Authorised operator and active destination exist; Enter transfer details and save	Unique transfer records source, destination, priority, date, carrier or vehicle and notes	FR-OUT-005; FR-OUT-007-008		
UAT-OUT-006	Search and add transfer item	Active products exist; Search by partial SKU or name and enter quantity	Matching items appear and a valid line can be added, changed or removed	FR-OUT-006		
UAT-OUT-007	Reject invalid transfer data	Invalid destination, product or quantity is available; Attempt to save transfer	Actionable errors display and no inventory changes	FR-OUT-007; TR-OUT-004		
UAT-OUT-008	FIFO allocation across batches	Multiple eligible batches with different received dates exist; Allocate transfer line	Oldest eligible received batch is allocated first	FR-OUT-009; TR-OUT-005		
UAT-OUT-009	Batch and location control	SKU and batch already occupy an approved location; Attempt conflicting allocation	Approved location is retained and conflicting assignment is blocked	FR-OUT-010-011		
UAT-OUT-010	Transfer shortage	Requested quantity exceeds eligible stock; Allocate transfer	Shortage is calculated, highlighted and dispatch is blocked	FR-OUT-012		

Test ID	Scenario	Precondition / action	Expected result	Requirement	Status	Comment
UAT-OUT-011	Hold amend or cancel transfer	Transfer is allocated but not dispatched; Enter reason and perform approved action	Status follows rule and allocation is retained, changed or released exactly once	FR-OUT-013; GEN-OUT-005		
UAT-OUT-012	Pick validation	Allocated transfer is ready for picking; Scan wrong location, SKU or batch then correct stock	Wrong scan is rejected and correct stock is confirmed	FR-OUT-014; TR-OUT-004		
UAT-OUT-013	Short-pick exception	Allocated stock cannot be fully picked; Record short quantity and reason	Exception remains visible and dispatch stays blocked pending resolution	FR-OUT-015		
UAT-OUT-014	Dispatch completed transfer	All lines are allocated and picked; Confirm dispatch	Reservations are consumed once, warehouse on hand reduces and transfer becomes in transit	FR-OUT-016		
UAT-OUT-015	Dispatch record and tracking	Transfer is ready for dispatch; Confirm dispatch and view record	Viewable dispatch record contains transfer, destination, vehicle or carrier, items, batches and tracking	FR-OUT-017		
UAT-OUT-016	Receive transfer at destination store	Transfer is in transit; Count and confirm full receipt	In-transit reduces and eligible store on hand and order available increase once	FR-OUT-018; FR-OUT-020		
UAT-OUT-017	Receipt variance	Received quantity or condition differs from expected; Record shortage, overage or damage	Variance and reason are recorded; affected quantity is not silently promoted	FR-OUT-019-020		
UAT-OUT-018	Role-based access	Users with different outbound roles exist; Attempt approved and restricted actions	Approved actions succeed and restricted actions are blocked without data change	FR-OUT-022; NFR-OUT-007; TR-OUT-009		
UAT-OUT-019	Outbound history and export	Completed transfers and orders exist; Filter history and export results	Screen and export contain the same filtered traceable records	FR-OUT-021; FR-OUT-025		
UAT-OUT-020	Integration outage and recovery	Outbound event delivery can be interrupted; Interrupt service, restore it and retry event	Event completes once, balances reconcile and no transaction is lost or duplicated	FR-OUT-023-024; NFR-OUT-002; NFR-OUT-006; TR-OUT-007		

12.1 Evidence gate

- All Must requirements have passed or have an approved exception with owner and due date.
- No open defect can create negative stock, duplicate stock movements or cross the Main Warehouse and store-order boundary.
- Warehouse, store and website balances reconcile for every pilot transaction.
- Operational users have approved the workflow, exception handling, history and printed or exported output.
- Rollback and recovery steps have been tested before broader store enablement.